(1) Solve $y'' - y' - 2y = 3e^{2x}$

Homo eqn $y'' - y' - 2y = 0$

Ch Eqn $r^2 - r - 2 = 0$ Roots of ch Eqn

Solution is $y_{hom} = c_1 e^{2x} + c_2 e^{-x}$

Non-Homo Eqn $y'' - y' - 2y = 0$ (method of undetermined coefficients)

$$\boxed{\begin{array}{l} \text{Trial solution :- } ke^{2x} \\ y = c_1 e^{2x} + c_2 e^{-x} + k' e^{2x} \\ y = (c_1 + k') e^{2x} + c_2 e^{-x} \end{array}} \quad X$$

Rule :- If trial solution is part of solution of homogeneous Eqn, then we replace trial solution $f(x)$ with $x^m f(x)$, where m is the least positive integer such that $x^m f(x)$ is Not part of solution of Homo Eqn.

Trial solution :- $y = kxe^{2x}$

$$y' = 2kxe^{2x} + ke^{2x}$$

$$y'' = 4ke^{2x} + 4kxe^{2x}$$

We have

$$3e^{2x} = y'' - y' - 2y = 4ke^{2x} + \cancel{4kxe^{2x}} - \cancel{2kxe^{2x}} - ke^{2x} - \cancel{2kxe^{2x}}$$

$$3e^{2x} = 3ke^{2x} \Rightarrow k = 1$$

Solution of Non hom Eqn containing No constant

P.I $\longrightarrow$ xe^{2x}

Complete solution

$$y = c_1 e^{2x} + c_2 e^{-x} + xe^{2x}$$

eqn containing No constant $\quad y = ce$

$D \rightarrow$ Differential operator. $\quad D = \frac{d}{dx}$

$p(D) \rightarrow$ polynomial in D

$\quad \hookrightarrow a_n D^n + a_{n-1} D^{n-1} + \_\_ + a_0$

$\quad \hookrightarrow$ Diff operator.

$$p(D)y = (a_n D^n + a_{n-1} D^{n-1} + \_\_ + a_0)y = a_n y^n + a_{n-1} y^{n-1} + \_\_ + a_0 y$$

* $\quad \frac{1}{D} f(x) = \int f(x)\,dx = g(x), \quad D(g(x)) = f(x)$

* $\quad \frac{1}{p(D)} f(x) \rightarrow$ a function of x, which returns $f(x)$, when it is operated by $p(D)$

$$\frac{1}{p(D)} f(x) = g(x), \quad \text{such that} \quad p(D)(g(x)) = f(x)$$

For Example :- $\quad \frac{1}{D^2+3D}\underset{f(x)}{(2+6x)} = \underset{g(x)}{x^2} \quad$ check!

$$(D^2+3D)(x^2) = 2+6x.$$

Diff Eqn $\quad p(D)\,y = r(x)$

P.I. $\quad y = \boxed{\frac{1}{p(D)} r(x)}$

Theorem:- If $r(x)$ is a function of x, then

Thm :- If $r(x)$ is a function of x, then

Rule I

$$\boxed{\frac{1}{D-\alpha} r(x) = e^{\alpha x} \int r(x)\, e^{-\alpha x}\, dx}$$

$$\frac{1}{D} r(x) = \int r(x)\, dx$$

$$\frac{1}{D-0} r(x) = \int r(x)\, dx$$

Proof :-

$$y = \frac{1}{(D-\alpha)} r(x)$$

$$D = \frac{d}{dx}$$

$$(D-\alpha) y = r(x)$$

$$\frac{dy}{dx} - \alpha y = r(x)$$

$$I.F = e^{\int -\alpha\, dx} = e^{-\alpha x}$$

Solution of linear Eqn

$$y \times e^{-\alpha x} = \int r(x)\, e^{-\alpha x}\, dx \Rightarrow y = e^{\alpha x} \int r(x)\, e^{-\alpha x}\, dx$$

Rule II

$$\frac{1}{f(D)} e^{\alpha x} = \frac{1}{f(\alpha)} e^{\alpha x} \quad \text{if} \quad \boxed{f(\alpha) \neq 0}$$

- $$\frac{1}{kD} e^{\alpha x} = \frac{1}{k} \int e^{\alpha x}\, dx = \frac{e^{\alpha x}}{k\alpha}$$
- $$\frac{1}{k' D^2} e^{\alpha x} = \frac{1}{k' D}\left(\frac{e^{\alpha x}}{\alpha}\right) = \frac{1}{k' \alpha} \int e^{\alpha x}\, dx = \frac{e^{\alpha x}}{k' \alpha^2}$$

Suppose $f(\alpha) = 0 \Rightarrow \alpha$ is root of polynomial $f(D)$, $\deg(f(D)) = n$

$$(D-\alpha)^n$$

$$(D-\alpha)^k \varphi(D), \quad \varphi(\alpha) \neq 0$$

$k < n$

$k > 1$

(1) $$\frac{1}{(D-\alpha)^n} e^{\alpha x} = \frac{x^n}{n!} e^{\alpha x}$$

$\boxed{n=1}$ $$\frac{1}{(D-\alpha)} e^{\alpha x} = x e^{\alpha x}$$ Check!

$$(D-\alpha)(x e^{\alpha x}) = \alpha x e^{\alpha x} + e^{\alpha x} - \alpha x e^{\alpha x}$$
$$= e^{\alpha x}$$

$\boxed{n=2}$ $$\frac{1}{(D-\alpha)^2} e^{\alpha x} = \frac{x^2}{2!} e^{\alpha x} = \frac{x^2}{2} e^{\alpha x}$$

Check $$(D-\alpha)^2 \left(\frac{x^2}{2} e^{\alpha x}\right) = e^{\alpha x}$$

$$\left(D^2 - 2\alpha D + \alpha^2\right)\left(\frac{x^2}{2} e^{\alpha x}\right)$$

Case II :- $$\frac{1}{f(D)} e^{\alpha x}, \quad f(D) = (D-\alpha)^k \varphi(D), \quad \varphi(\alpha) \neq 0.$$

$$\frac{1}{f(D)} e^{\alpha x} = \frac{1}{(D-\alpha)^k \varphi(D)} e^{\alpha x}$$

$$= \frac{1}{(D-\alpha)^k}\left(\frac{1}{\varphi(D)} e^{\alpha x}\right) = \frac{1}{(D-\alpha)^k} \cdot \frac{1}{\varphi(\alpha)} e^{\alpha x}$$

$$= \frac{1}{\varphi(\alpha)}\left(\frac{1}{(D-\alpha)^k} e^{\alpha x}\right) = \frac{1}{\varphi(\alpha)} \cdot \frac{x^k}{k!} e^{\alpha x}$$

Example :- $$(D+2)(D-1)^3 y = e^x$$

Solution :- P.I. $$\frac{1}{(D+2)(D-1)^3} e^x = \frac{1}{3} \cdot \frac{x^3}{3!} e^x$$

(2) $$\frac{1}{f(D^2)} \sin(ax) \quad \text{or} \quad \frac{1}{f(D^2)} \cos(ax)$$

(2) $\frac{1}{f(D)}\sin(ax)$ or $\frac{1}{f(D)}\cos(ax)$

- $$\boxed{\frac{1}{\varphi(D^2)}\sin(ax) = \frac{1}{\varphi(-a^2)}\sin(ax)}$$ if $\varphi(-a^2) \neq 0$

- $$\frac{1}{\varphi(D^2)}\cos(ax) = \frac{1}{\varphi(-a^2)}\cos(ax)$$ if $\varphi(-a^2) \neq 0$

$y = \sin ax$, $y' = a\cos ax$, $y'' = -a^2 \sin ax$

Example :- $(D^2 - 3D + 2)y = \sin 3x$

P.I. $$\frac{1}{D^2 - 3D + 2}\sin 3x = \frac{1}{-9 - 3D + 2}\sin 3x$$

$$= \frac{1}{-3D - 7}\sin 3x$$

$$= \frac{-1\,(3D-7)}{(3D+7)(3D-7)}\sin 3x$$

$$= \frac{7 - 3D}{9D^2 - 49}\sin 3x = \frac{7 - 3D}{-81 - 49}(\sin 3x)$$

$$= \frac{1}{-130}\left(7\sin 3x - 9\cos 3x\right)$$

P.I. $\frac{1}{f(D)}x^m$ or $\frac{1}{f(D)}p(x) \longrightarrow$ polynomial in x.

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \text{———}$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - x^5 + \text{———}$$

$$\cdot\ \frac{1}{f(D)}\,p(x) = \frac{1}{(1+\varphi(D))}\,p(x) = (1+\varphi(D))^{-1}\,(p(x))$$

Example :- $\quad (D^2 + D)\,y = x^2 + 2x + 4$

P.I. $\quad \dfrac{1}{D^2+D}\,(x^2+2x+4)$

$$= \frac{1}{D(D+1)}\,(x^2+2x+4) = \frac{1}{D}\,(1+D)^{-1}\,(x^2+2x+4)$$

$$= \frac{1}{D}\left(1 - D + D^2 - D^3 + \text{———}\right)(x^2+2x+4)$$

$$= \frac{1}{D}\left(x^2 + \cancel{2x} + 4 - \cancel{2x} - \cancel{2} + \cancel{2}\right) = \frac{1}{D}\,(x^2+4)$$

$$= \int (x^2+4)\,dx = \frac{x^3}{3} + 4x.$$

$$\boxed{\frac{1}{D-\alpha}\,r(x) = e^{\alpha x}\int r(x)\,e^{-\alpha x}\,dx}$$

* $\quad (D^2 + a^2)\,y = \tan(ax)$

P.I. $\quad \dfrac{1}{(D^2+a^2)}\,\tan(ax)$

$$= \frac{1}{\text{———}}\ \tan(ax)$$

$$(D+ai)(D-ai)$$

$$= \left(\frac{A}{D+ai} + \frac{B}{D-ai}\right)(\tan ax)$$